

Algebraic Fractions – Word Problems

Worksheet 1 (Calculator)

1. W11/qp22/q10

A piece of wire, 28 cm in length, is cut into two parts. One part is used to make a rectangle and the other a square. The length of the rectangle is three times its width. The width of the rectangle is x centimetres.



- a.
 - i. Write down an expression, in terms of x , for the length of the rectangle. [1]
 - ii. Find, and simplify, an expression, in terms of x , for the length of a side of the square. [2]
- b. It is given that the area of the rectangle is equal to the area of the square.
 - i. Form an equation in x and show that it reduces to $x^2 - 28x + 49 = 0$. [2]
 - ii. Solve the equation $x^2 - 28x + 49 = 0$, giving each solution correct to 3 significant figures. [4]
 - iii. Which solution represents the width of the rectangle? Give a reason for your answer. [2]
 - iv. Calculate the area of the square. [1]

2. W13/qp22/q3(d)

The sum of three consecutive integers is 84. Find these three integers. [2]

3. S14/qp21/q11

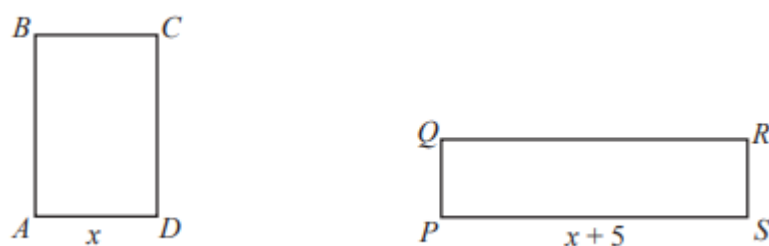
Imran drives a distance of 180 km on a business trip. He drives the first 100km at an average speed of x km/h. He drives at an average speed 5 km/h slower than this for the remainder of the journey.

- Find, in terms of x , an expression for the time taken, in hours, for the first 100 km. [1]
- Given that the journey takes a total of 2 hours 30 minutes, form an equation in x and show that it simplifies to $x^2 - 77x + 200 = 0$. [4]
- Solve the equation $x^2 - 77x + 200 = 0$, giving each answer correct to 2 decimal places. [4]
- Which of the solutions in part (c) represents the speed for the first 100km of Imran's trip? Give a reason for rejecting the other solution. [1]
- Find the difference between the times taken for the first and second parts of the journey. Give your answer in minutes, correct to the nearest minute. [2]

4. W16/qp21/q2(f)

Abebi, Bella and Chuku share \$112. Abebi receives $\$x$. Bella receives \$12 less than Abebi. Chuku receives twice as much as Bella. Form an equation in x and solve it to find how much Chuku receives. [3]

5. W16/qp22/q7(b)



ABCD and PQRS are rectangles. $AD = x$ cm and $PS = (x + 5)$ cm. Each rectangle has an area of 17cm^2 .

- Write down an expression for PQ in terms of x . [1]
- AB is 3 cm longer than PQ. Form an equation in x and show that it simplifies to $3x^2 + 15x - 85$. [3]
- Solve the equation $3x^2 + 15x - 85$. Give your solutions correct to 3 significant figures. [3]
- Find the perimeter of the rectangle PQRS. [2]

6. W17/qp21/q9(b)

A tank holds 2500 litres of oil. A small pump can add oil to the tank at a rate of x litres per minute. A large pump can add oil to the tank at a rate of $(x + 20)$ litres per minute.

- i. Write down an expression, in terms of x , for the number of minutes the small pump takes to fill the empty tank. [1]
- ii. It takes 15 minutes longer to fill the empty tank using the small pump than it does with the large pump. Form an equation in x and show that it simplifies to $3x^2 + 60x - 10\,000 = 0$. [3]
- iii. Solve the equation $3x^2 + 60x - 10\,000 = 0$. Give each answer correct to 2 decimal places. [3]
- iv. Find the length of time the large pump takes to fill the empty tank. Give your answer in minutes and seconds, correct to the nearest second. [3]

7. W20/qp22/q5

- a. Gita has n stamps. Ravi has twice as many stamps as Gita. Sanjay has 7 fewer stamps than Ravi. Altogether, the three children have 108 stamps. Form an equation in n and solve it to find the number of stamps Sanjay has. [3]
- b. Simplify $\frac{6t^2 2v^3}{5} \div \frac{3t^2}{v^2}$ [2]
- c. Simplify $\frac{x^2 - 16}{3x^2 + 10x - 8}$ [3]

8. W20/qp22/q10

Amira drives 40 km to work.

- a. Amira takes x minutes to drive the first 30km of the journey. Show that her average speed in km/h for the first 30km of the journey is $\frac{1800}{x}$. [1]
- b. Amira's average speed in km/h for the final 10km of the journey is $\frac{600}{x-25}$. Her average speed for the first 30km of the journey is 8km/h slower than her average speed for the final 10km. Form an equation in x and show that it simplifies to $x^2 + 125x - 5625$. [3]
- c. Solve the equation $x^2 + 125x - 5625$. Show your working and give each answer correct to 1 decimal place. [3]
- d. It takes Amira 25 minutes less to drive the final 10km than it takes for the first 30km. Calculate Amira's average speed, in km/h , for the whole journey. [3]

Answer Key:

1. (a)(i) $3x$ (a)(ii) $7-2x$ (b)(i) Shown (ii) 1.88 and 26.1 (iii) 1.88 with convincing reason (iv) 10.6 or 10.5

2. 27, 28, 29

3. (a) $\frac{100}{x}$ (b) Shown (c) 74.31 (d) 74.31 because 2.69 would give negative speed for second part. (e) 11

4. 50

5. (i) $PQ = \frac{17}{x+5}$ (ii) Shown (iii) 3.38 -8.38 (iv) 20.8

6. (i) $\frac{2500}{x}$ (ii) Shown (ii) 48.59 and -68.59 (iv) 36 minutes 27 seconds

7. (a) 39 (b) $\frac{2v^5}{5}$ (c) $\frac{x-4}{3x-2}$

8. (a) $\frac{30}{x} \times 60 = \frac{1800}{x}$ (b) Shown (c) 53.0 to 53.1